

VIEW

-----VIRTUAL TABLE-----

--1 CREATE VIEW THAT SHOW FIRSTNAME, LASTNAME AND SALARY
--(COLUMN FILTER)

```
CREATE VIEW VW_NAME_SAL
AS
SELECT FIRSTNAME, LASTNAME, SALARY
FROM EMPLOYEES
```

```
SELECT * FROM VW_NAME_SAL
```

```
drop view VW_NAME_SAL
--2 CREATE VIEW OF IT DEPRATMENT EMPLOYEE ONLY
--(ROW FILTER)
```

```
CREATE VIEW VW_IT_EMP
AS
SELECT *
FROM EMPLOYEES
WHERE DEPARTMENT='IT'
```

```
SELECT * FROM VW_IT_EMP
```

```
drop view VW_IT_EMP
---3 CREATE VIEW THAT SHOWS FULL NAME AND SALARY OF HR DEPT
```

```
CREATE VIEW VW_FULLNAME_SALARY_HR
AS
SELECT FIRSTNAME+LASTNAME AS FULLNAME, SALARY
FROM EMPLOYEES
WHERE DEPARTMENT='HR'
```

```
SELECT * FROM VW_FULLNAME_SALARY_HR
```

```
drop view VW_FULLNAME_SALARY_HR
--4 CREATE A VIEW THAT SHOWS THE AVERAGE SALARY OF EVERY DEPARTMENT
```

```
CREATE VIEW VW_AVG_SAL_DEPT
AS
SELECT DEPARTMENT, AVG(SALARY) AS AVG
FROM EMPLOYEES
GROUP BY DEPARTMENT
```

```
SELECT * FROM VW_AVG_SAL_DEPT
```

```
drop view VW_AVG_SAL_DEPT
--5 CREATE VIEW THAT SHOWS THE ANNUAL SALARY OF EMPLOYEE
```

```
CREATE VIEW VW_ANU_SALARY
AS
SELECT FIRSTNAME, SALARY*12 AS ANNUAL_SAL
FROM EMPLOYEES
```

```
SELECT * FROM VW_ANU_SALARY
drop view VW_ANU_SALARY
--6 CREATE VIEW OF THOSE EMP WHOSE NAME STARTS WITH 'J'
--AND SALARY IS MORE THEN 50000
```

```
CREATE VIEW VW_NAME_PATTERN
AS
SELECT FIRSTNAME
FROM EMPLOYEES
WHERE FIRSTNAME LIKE 'J%' AND SALARY>50000

SELECT * FROM VW_NAME_PATTERN
drop view VW_NAME_PATTERN
--7 CREATE VIEW WITH ID,FIRSTNAME, DEPARTMENT,SALARY

CREATE VIEW VW_EMPVIEW
AS
SELECT EMPLOYEEID,FIRSTNAME,DEPARTMENT,SALARY
FROM
EMPLOYEES

SELECT * FROM VW_EMPVIEW

drop view VW_EMPVIEW
-----INSERT IN VIEW-----
SELECT * FROM VW_EMPVIEW

--8 INSERT RECORD IN VIEW
INSERT INTO VW_EMPVIEW VALUES(10, 'MILI', 'IT', 60000)

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES

-----UPDATE THE VIEW-----

--9 UPDATE DEPT TO 'HR' IN VIEW WHOSE ID IS 10
UPDATE VW_EMPVIEW
SET DEPARTMENT = 'HR'
WHERE EMPLOYEEID=10

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES

-----DELETE FROM THE VIEW-----

--10 DELETE RECORD FROM VIEW

DELETE FROM VW_EMPVIEW
WHERE FIRSTNAME='MILI'

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES

----AGGREGATE FUNCTIONS, GROUP BY, HAVING CLAUSE IN VIEW----

--11 CREATE VIEW TO DISPLAY SUM OF SALARY FOR EACH DEPT

CREATE VIEW VW_SUM_SAL_DEPT
```

```
AS
SELECT DEPARTMENT,SUM(SALARY) AS TOTAL
FROM EMPLOYEES
GROUP BY DEPARTMENT

SELECT * FROM VW_SUM_SAL_DEPT

drop view VW_SUM_SAL_DEPT
--ERROR
UPDATE VW_SUM_SAL_DEPT SET
TOTAL=10000
WHERE DEPARTMENT='HR'

--WHY ERROR?
--view isn't updatable
--it's an aggregate view that uses SUM() and GROUP BY.
--In SQL, views created with aggregation or grouping are read-only

-----CREATE VIEW WITH DISTINCT-----
SELECT * FROM EMPLOYEES
--12 VIEW WITH DISTINCT

CREATE VIEW VW_DISTINCTEMP
AS
SELECT DISTINCT FIRSTNAME
FROM
EMPLOYEES

SELECT * FROM VW_DISTINCTEMP

drop view VW_DISTINCTEMP
--ERROR (DUE TO DISTINCT)
--View uses DISTINCT, which makes it non-updatable.

INSERT INTO VW_DISTINCTEMP VALUES('KRITI');

--ERROR
UPDATE VW_DISTINCTEMP
SET FIRSTNAME='JOHN' WHERE
FIRSTNAME='EMILY'

-----VIEW ON BASE TABLE WITH PK-----
drop table STU_INFO

CREATE TABLE STU_INFO
(ID INT PRIMARY KEY,
NAME VARCHAR(20),
SPI DECIMAL(4,2));

INSERT INTO STU_INFO VALUES(1,'MITA',5);
INSERT INTO STU_INFO VALUES(2,'RINA',7);
INSERT INTO STU_INFO VALUES(3,'KRISHNA',9);
INSERT INTO STU_INFO VALUES(4,'KRISHNA',10);

SELECT * FROM STU_INFO

--14 VIEW ON BASETABLE WITH PK
CREATE VIEW VW_NAME_SPI
AS
```

```
SELECT NAME,SPI FROM
STU_INFO;
```

```
SELECT * FROM VW_NAME_SPI
```

```
drop view VW_NAME_SPI
--ERROR (AS BASE TABLE CONTAINS PK)
```

```
INSERT INTO VW_NAME_SPI VALUES('MITAL',8);
```

```
-----COMPLEX VIEW-----
```

```
CREATE TABLE STUDENT_MASTER(
    STUDENTID INT PRIMARY KEY,
    STUDENTNAME VARCHAR(255) NOT NULL,
);
```

```
INSERT INTO STUDENT_MASTER(STUDENTID, STUDENTNAME) VALUES
(1, 'JOHN DOE'),
(2, 'JANE SMITH'),
(3, 'EMILY DAVIS');
```

```
CREATE TABLE COURSES(
    COURSEID INT PRIMARY KEY,
    COURSENAME VARCHAR(255) NOT NULL,
    STUDENTID INT FOREIGN KEY (STUDENTID) REFERENCES STUDENT_MASTER(STUDENTID)
);
```

```
INSERT INTO COURSES (COURSEID, COURSENAME, STUDENTID ) VALUES
(101, 'MATHEMATICS', 1),
(102, 'PHYSICS', 2),
(103, 'LITERATURE', 1),
(104, 'MATHEMATICS', 3);
```

```
select * from STUDENT_MASTER
select * from COURSES
```

```
---15 VIEW WITH MULTIPLE TABLE
```

```
CREATE VIEW VW_COMPLEX
AS
SELECT STUDENT_MASTER.STUDENTNAME,COURSES.COURSENAME
FROM STUDENT_MASTER JOIN
COURSES
ON STUDENT_MASTER.STUDENTID=COURSES.STUDENTID
```

```
SELECT * FROM VW_COMPLEX
```

```
drop view VW_COMPLEX
---ERROR
--is not updatable because the modification affects multiple base tables.
```

```
INSERT INTO VW_COMPLEX VALUES('MEERA','MATHS')
```

```
DELETE FROM VW_COMPLEX WHERE STUDENTNAME='JOHN DOE'
```

```
-----ALTER VIEW-----
```

```
CREATE VIEW VW_EMP
AS
SELECT EMPLOYEEID,FIRSTNAME
FROM
EMPLOYEES
```

```
SELECT * FROM VW_EMP

----16 MODIFY THE EXISTING VIEW WITH ALTER
ALTER VIEW VW_EMP
AS
SELECT EMPLOYEEID, FIRSTNAME, SALARY
FROM
EMPLOYEES

drop view VW_EMP
---17 RENAME VIEW
SP_RENAME VW_EMP, VW_EMP_NEW

SELECT * FROM VW_EMP_NEW

--18 DROP VIEW
DROP VIEW VW_EMP_NEW

--19 READ ONLY VIEW--

CREATE VIEW VW_EMP_DATA
AS
SELECT EMPLOYEEID, FIRSTNAME, SALARY
FROM
EMPLOYEES

SELECT * FROM VW_EMP_DATA
SELECT * FROM EMPLOYEES

INSERT INTO VW_EMP_DATA VALUES(25, 'HINA', 25000);

--NOW MAKE THE VIEW READ ONLY

CREATE TRIGGER TR_VIEW
ON VW_EMP_DATA
INSTEAD OF INSERT, UPDATE, DELETE
AS
BEGIN
    PRINT 'YOU ARE NOT ALLOWED TO PERFORM DML OPERATION'
END

--NOW INSERT IN VIEW
INSERT INTO VW_EMP_DATA VALUES(30, 'HEMAL', 25000);

SELECT * FROM VW_EMP_DATA
SELECT * FROM EMPLOYEES

UPDATE VW_EMP_DATA SET SALARY=100
WHERE EMPLOYEEID=2

DELETE FROM VW_EMP_DATA WHERE EMPLOYEEID=2
```